Isabel Brindley

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Summary

Master's student in Mechatronics, Robotics and Biomechanical Engineering at the Technical University of Munich. Completed a BSc in Physics (Mathematics minor) from Monash University, including an exchange semester at the University of Groningen. Hands-on experience across power electronics, battery systems, hardware prototyping and CAD, gained through a startup developing an AI wearable for elderly care, a multidisciplinary student engineering team and a professional engineering internship. Strong foundation in experimental design, programming and technical writing, with experience leading and coordinating teams. Seeking to apply this to high-tech robotics, with a particular interest in biomechatronics and impact-driven applications such as caregiving.

Education

Technical University of Munich Oct. 2026 – Present
MSc. Mechatronics, Robotics and Biomechanical Engineering *Munich, Germany*

- Pursuing topics in Robotics, Control Theory, AI, Dynamics, Programming and Biomechanical engineering to become a competent all-rounder engineer.
- Selected for the Robo-Dodo lab course, working on simulation and control of a bipedal walking robot.

Monash University Jan. 2023 – Jul. 2026
Bachelor of Science (Physics) *Melbourne, Australia*

- WAM: 81/100 | GPA: 3.4/4.0
- Areas of study: Physics, Mechatronics Engineering, Computer Science, Mathematics, Astronomy
- Undertook a research project under the supervision of Dr. Mark Edmonds in materials physics. In my project, titled *Understanding the growth and electronic properties of rare earth Kagome metals*, I grew thin film samples of rare-earth Kagome metal $TbMn_6Sn_6$ with Molecular Beam Epitaxy and analysed their chemical composition and electrical properties using various Photoemission Spectroscopy techniques.

Rijksuniversiteit Groningen (Academic Exchange) Jan. 2025 – Jun. 2025
Bachelor of Science *Groningen, Netherlands*

- Coursework in Particle Physics, Mathematics and Astronomy.

Melbourne Girls Grammar Graduated Nov. 2022
VCE *Melbourne, Australia*

- 99.65/100.00 ATAR

Professional Experience

Electrical Engineer (Student Team) Mar. 2026 – Jun. 2026
MHAB *Melbourne, Australia*

During my time with MHAB, a multi-disciplinary student team, I contributed to the power engineering sub-division in developing a flight-ready product. I designed LTspice schematics, including a buck converter, and researched battery configurations for low-pressure, low-temperature environments, weighing safety, cost and efficiency. I also developed battery safety testing experiments, applying physics principles to the risks of operating in extreme conditions.

Engineer (Hardware & Product Development) Nov. 2025 – Mar. 2026
Idiom *Melbourne, Australia*

Worked at the forefront of Idiom, an early-stage startup developing an AI wearable for elderly care. I analysed AI software codebases, researched battery configurations and designed a prototype shell. I also explored the market with the founder and discussed business ideas, contributing to early product and business direction.

Engineering Internship May 2024
Traffix Group *Melbourne, Australia*

To explore a career pathway in Civil Engineering, I undertook an internship with Traffix Group, a traffic engineering consultancy. I designed carparks and developed swept paths using CAD software. I conducted on-site traffic surveys for new housing developments and contributed to traffic reports for proposed developments and infrastructure projects.

Outreach Manager (Student Team) Mar. 2023 – Jun. 2024
Precious Plastics Monash *Melbourne, Australia*

Starting as a team-member and then working up to manager, I was part of a multidisciplinary team of eight students, promoting sustainability and recycling through engineering-based workshops. I delivered presentations to primary, secondary and university audiences, while planning events, managing budgets and liaising with stakeholders across the Monash community.

Other Experience

Store Assistant, ALDI Oct. 2023 – Nov. 2025
Waitress, North Point Cafe Nov. 2021 – Jan. 2023
Team Member, Grill'd Jun. 2020 – Nov. 2021

Skills & Interests

Programming: Python, C, C++, MATLAB, Wolfram Language, LaTeX
Hardware & Embedded: Verilog, MIPS Assembly, Arduino
Software: SolidWorks, Microsoft 365
Languages: English (Native), French (B2)
Interests: Bouldering, Hiking, Travelling, Reading

References

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Dr. Mark Edmonds from School of Physics and Astronomy Monash University
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